

WASP-10b

R-0827

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_250908 · created 2026-07-28T16:33:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460926.817 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.047
Transit depth (Rp/Rs)^2	3.62 +/- 1.78 [%]
Orbital Inclination (inc)	87.46 +/- 1.42
Airmass coefficient 1 (a1)	0.9 +/- 0.03
Airmass coefficient 2 (a2)	0.1 +/- 0.24
Scatter in the residuals of the lightcurve fit is	3.27 %
Best Comparison Star	None
Optimal Aperture	3.15
Optimal Annulus	6.48
Transit Duration (day)	0.084 +/- 0.016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.047 vs 0.1592 (deviation 0.66σ)
Duration fit vs published	0.084 d vs 0.0946 d (deviation 0.66σ)
Mid-time O–C	7.0 min
Depth SNR	4.0
Residual scatter	3.27 %

Light curve

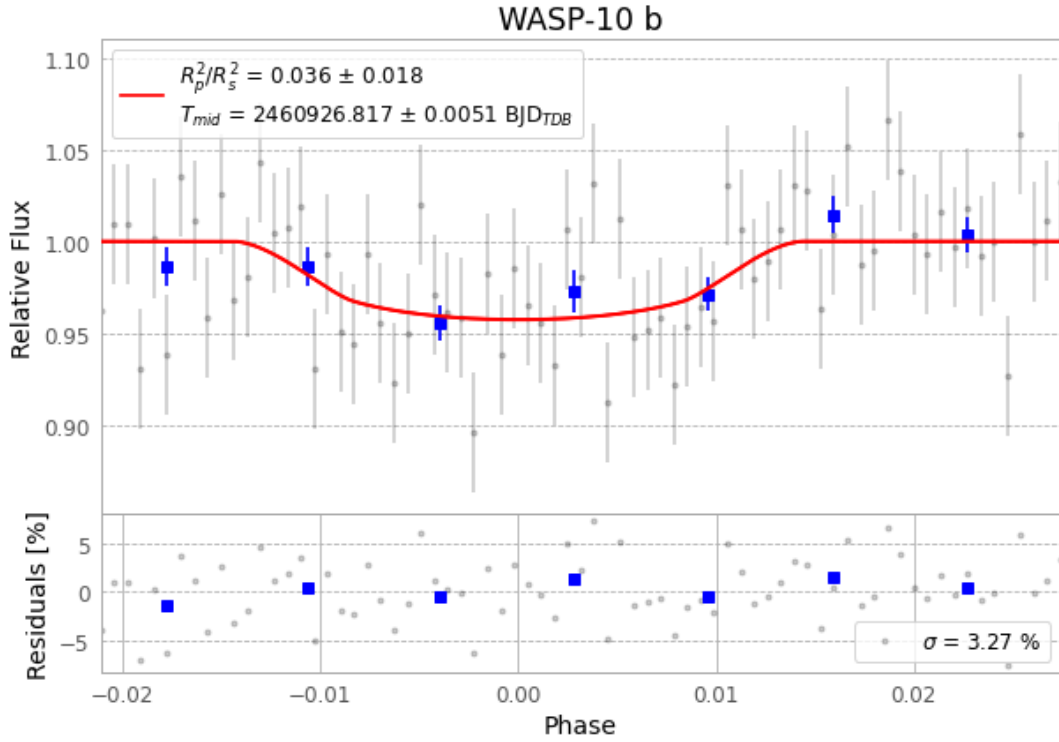


Figure 1 — FinalLightCurve_WASP-10 b_2025-09-07.png (SHA-256 16843080b6198a74...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [294, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50c32fe4b679cb34...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

73 FITS frames (48 MB), WASP-10250908055416.FITS → WASP-10250908093015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-250908.log (d348b344cb7c...), FinalLightCurve_WASP-10 b_2025-09-07.png (16843080b619...), FinalLightCurve_WASP-10 b_2025-09-07.csv (55d2de43325d...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 16:33Z.